

4(a)(i)	caused by pathogens ; is transmissible / AW ;	2
4(a)(ii)	<i>Mycobacterium tuberculosis</i> / <i>Mycobacterium bovis</i> ;	1
4(b)	natural and passive ;	1
4(c)	any three from: 1 (as influenza virus mutates) antigen (proteins) altered / altered capsid proteins / AW ; 2 antibodies are specific to antigens ; 3 ref. to complementary shape (between variable regions of antibody and antigen) ; 4 ref. to variable regions / Fab, at the ends ;	3
4(d)(i)	<i>for children, allow people</i> <i>for polio allow disease</i> any three from: 1 ref. to vaccination programme aims to vaccinate, whole populations / large numbers ; 2 vaccine allows (most), children / people, to have an immune response (against antigens of the disease organism) ; 3 develop, long-term immunity / artificial active immunity / memory cells against, polio / disease ; 4 <i>idea that</i> vaccinated individuals cannot, transmit virus / transmit polio / infect others ; 5 herd immunity / described ; 6 AVP ; e.g. ref. to vaccination programme aims to break the transmission cycle	3

Question	Answer	Marks
4(d)(ii)	any one from: penicillin <u>only</u> acts on, bacteria / prokaryotes ; virus does not have, cell walls / murein / peptidoglycan ; virus does not have, transpeptidases / the enzyme that is, inhibited / acted on, by penicillin ; virus does not grow / penicillin (only) acts on growing cells / AW ; A when cell wall is, growing (larger) for growing cells virus does not have cellular structure / virus is acellular ;	1

Question	Answer	Marks
----------	--------	-------